

## Discrete Structure

## Section 3

## Assignment 2

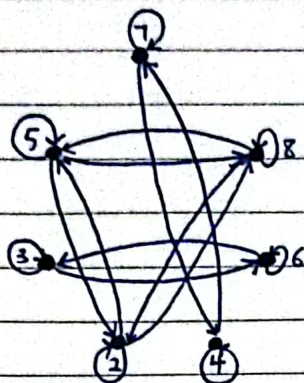
|        |                  |           |
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## Assignment 2

Q1



$$xRy \iff x-y=3n, n \in \mathbb{Z}$$

$$R = \{(2,2), (2,5), (2,8), (3,3), (3,6), (4,4), (4,7), (5,2), (5,5), (5,8), (6,3), (6,6), (7,4), (7,7), (8,2), (8,5), (8,8)\}$$

Q2

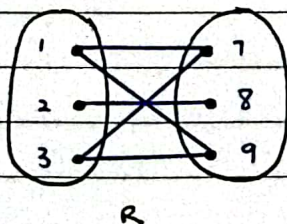
$$A = \{1, 2, 3\}, B = \{9, 8, 7\}$$

$$aRb \iff a+b = \text{even}, (a,b) \in A \times B$$

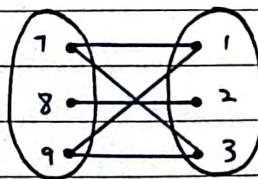
$$a) R = \{(1,9), (1,7), (2,8), (3,9), (3,7)\}$$

$$R^{-1} = \{(9,1), (7,1), (8,2), (9,3), (7,3)\}$$

b)



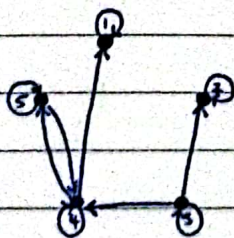
R

 $R^{-1}$ 

c)  $R^{-1}$  is the reverse function of  $R$  if  $R$  : relation  $A$  to  $B$ ,  
 $R^{-1}$  : relation  $B$  to  $A$

$$R^{-1} = \text{From } B \text{ to } A, bRa \iff b+a = \text{even}, (b,a) \in B \times A$$

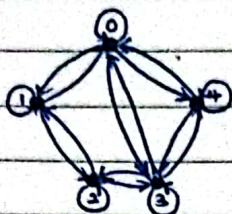
Q3



| domain | $\Sigma$ in degree | $\Sigma$ out degree |
|--------|--------------------|---------------------|
| 1      | 2                  | 1                   |
| 2      | 2                  | 1                   |
| 3      | 1                  | 3                   |
| 4      | 3                  | 3                   |
| 5      | 2                  | 2                   |



Q4



- it is reflexive,  $\forall x \in A, (x, x) \in R$
- it is symmetric,  $\forall a, b \in A, (a, b) \in R \rightarrow (b, a) \in R$
- it is not transitive, there exist  $(0, 1) \in R, (1, 2) \in R$ , but there is no  $(0, 2)$  in the Relation set.

Q5

$$R = \{(x, y) : 3x - y = 0\} \quad A = \{1, 2, 3, \dots, 14\}$$

$$R = \{(1, 3), (2, 6), (3, 9), (4, 12)\}$$

- a) It is not reflexive, because for  $x=1, y=1$  doesn't exist in the R set.

$$\text{Reflexive} : \forall x \in A, (x, x) \in R$$

- b) It is not symmetric, as a example  $(1, 3) \in R$ , but  $(3, 1) \notin R$

$$\text{Symmetric} : \forall a, b \in A, (x, y) \in R \rightarrow (y, x) \in R$$

- c) It is not transitive,  $(1, 3) \in R, (3, 9) \in R$  but  $(1, 9) \notin R$

$$\text{Transitive} : \forall a, b \in A, (a, b) \in R \wedge (b, c) \in R \rightarrow (a, c) \in R$$

Q6

R  $\otimes$  S

$$\begin{bmatrix} 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \otimes \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

$$(1, 1) \quad (0 \wedge 1) \vee (0 \wedge 0) \vee (1 \wedge 0) \vee (1 \wedge 0) = 0$$

$$(1, 2) \quad (0 \wedge 0) \vee (0 \wedge 1) \vee (1 \wedge 1) \vee (1 \wedge 0) = 1$$

$$(1, 3) \quad (0 \wedge 0) \vee (0 \wedge 0) \vee (1 \wedge 1) \vee (1 \wedge 1) = 1$$

$$(1, 4) \quad (0 \wedge 1) \vee (0 \wedge 1) \vee (1 \wedge 0) \vee (1 \wedge 1) = 1$$



$$\begin{array}{c} S \\ \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix} \end{array} \otimes \begin{array}{c} R \\ \begin{bmatrix} 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{array} = \begin{array}{c} \\ \begin{bmatrix} 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix} \end{array}$$

$$(1,1) \quad (1 \wedge 0) \vee (0 \wedge 1) \vee (0 \wedge 0) \vee (1 \wedge 0) = 0$$

$$(1,2) \quad (1 \wedge 0) \vee (0 \wedge 1) \vee (0 \wedge 0) \vee (1 \wedge 0) = 0$$

$$(1,3) \quad (1 \wedge 1) \vee (0 \wedge 0) \vee (0 \wedge 1) \vee (1 \wedge 0) = 1$$

$$(1,4) \quad (1 \wedge 1) \vee (0 \wedge 0) \vee (0 \wedge 1) \vee (1 \wedge 1) = 1$$

Q7) Function is a subset of relation

- In function, for each domain, there must only one value of codomain

- For relation, it can exist that domain didn't assigned to any codomain or assigned more than 1 codomain

Q8) i)  $\{(2,3), (3,4), (4,5), (5,2)\}$

= it is a function, each domain have assigned only 1 value.

ii)  $\{(2,4), (3,4), (5,4), (4,4)\}$

= it is a function, each domain have assigned only 1 value.

iii)  $\{(2,3), (2,4), (5,4)\}$

= it is not a function, there is 2 domains (3 and 4)

didn't assigned value. Besides, domain 2 has assigned to more than 1 value.



iv)  $\{(2,3), (3,5), (4,5)\}$

= it is not a function, since there is a domain (5) didn't assigned value

v)  $\{(1,2), (2,3), (4,4), (4,5)\}$

= it is not a function, there is 2 domains (3 and 5) didn't assigned value. And domain 2 and 4 is assigned to more than 1 value.

(Q9) Roster Form,  $R = \{(1,6), (2,7), (3,8), (4,9), (5,10)\}$

Domain =  $\{1, 2, 3, 4, 5\}$

Range =  $\{6, 7, 8, 9, 10\}$

(Q10) v)  $f: R \rightarrow R, f(x) = 1-2x$

$f(1) = 1-2(1)$   
 $= -1$

- It is one to one function, since for every variable  $f(x_1) \neq f(x_2)$

$f(2) = 1-2(2)$   
 $= -3$

- All codomain (Real num) have arrow connected from domain. Thus it is onto function.

$f(3) = 1-2(3)$   
 $= -5$

- It is bijective, since it is one to one and also onto.

vi)  $f: R \rightarrow R, f(x) = 5x^2 - 1$

$f(2) = 5(2)^2 - 1 = 19$

- It is not one to one function, since there exist  $f(x_1) = f(x_2)$

$f(-2) = 5(-2)^2 - 1 = 19$

range =  $[-1, +\infty)$

- It is not onto function, because not all codomain will be connected by domain. Range =  $[-1, +\infty)$  only.

- It is not bijective, because it is not both one to one and onto function.





vii)  $f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^4$

$$f(2) = 2^4$$

$$= 16$$

$$f(-2) = (-2)^4$$

$$= 16$$

- It is not one to one function, since exist  $f(x_1) = f(x_2)$

- It is not onto function, because not all codomain will be connected by domain. Range  $[0, +\infty)$  only

- It is not bijective since not both one to one and onto function.

viii)  $f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = \left(\frac{x-2}{x-3}\right)$

$$f(1) = \frac{1}{2}$$

$$f(2) = 0$$

$$f(-1) = -\frac{1}{2}$$

$$f(-2) = \frac{4}{5}$$

- It is one to one function, since for every variable  $f(x_1) \neq f(x_2)$

- It is onto function, because all the codomain will be connected from domain.

- It is bijective, because it is both one to one and onto function.

(Qii) ix)  $f(x) = 3x - 1; g(x) = x^2 - 1$

$$f(g(x)) = f(x^2 - 1)$$

$$= 3(x^2 - 1) - 1$$

0

$f(g(x))$

-4

1

-1

2

8

3

23

x)  $f(x) = x^2; g(x) = 5x - 6$

$$f(g(x)) = f(5x - 6)$$

$$= (5x - 6)^2$$

0

$f(g(x))$

36

1

1

2

16

3

81

xi)  $f(x) = x - 1; g(x) = x^3 + 1$

$$f(g(x)) = f(x^3 + 1)$$

$$= (x^3 + 1) - 1$$

$$= x^3$$

0

$f(g(x))$

0

1

1

2

8

3

27





(Q12)

$$\text{xii) } a_n = 6a_{n-1} - 9a_{n-2} ; a_0 = 1, a_1 = 6$$

$$a_0 = 1$$

$$a_1 = 6$$

$$a_2 = 6(6) - 9(1)$$

$$= 27$$

$$a_3 = 6(27) - 9(6)$$

$$= 108$$

$$a_4 = 6(108) - 9(27)$$

$$= 405$$

$$a_5 = 6(405) - 9(108)$$

$$= 1458$$

$$\text{xii) } a_n = 6a_{n-1} - 11a_{n-2} + 6a_{n-3} ; a_0 = 2, a_1 = 5, a_2 = 15$$

$$a_0 = 2$$

$$a_1 = 5$$

$$a_2 = 15$$

$$a_3 = 6(15) - 11(5) + 6(2)$$

$$= 47$$

$$a_4 = 6(47) - 11(15) + 6(5)$$

$$= 147$$

$$a_5 = 6(147) - 11(47) + 6(15)$$

$$= 455$$

$$a_6 = 6(455) - 11(147) + 6(47)$$

$$= 1395$$



$$\text{xiv) } a_n = -3a_{n-1} - 3a_{n-2} + a_{n-3} ; a_0 = 1, a_1 = -2, a_2 = -1$$

$$a_0 = 1$$

$$a_1 = -2$$

$$a_2 = -1$$

$$a_3 = -3(-1) - 3(-2) + 1 \\ = 10$$

$$a_4 = -3(10) - 3(-1) + (-2) \\ = -29$$

$$a_5 = -3(-29) - 3(10) + (-1) \\ = 56$$

$$a_6 = -3(56) - 3(-29) + 10 \\ = -71$$

$$\textcircled{\text{Q13}} \text{ i) } a_{n+1} = 5a_n - 3 ; a_1 = k$$

$$a_1 = k$$

$$a_2 = 5k - 3$$

$$a_3 = 5(5k - 3) - 3 \\ = 25k - 18$$

$$a_4 = 5(25k - 18) - 3 \\ = 125k - 93 \neq$$

$$\text{ii) } a_4 = 7$$

$$125k - 93 = 7$$

$$k = \frac{4}{5} \neq$$